

Friction Models and Friction Compensation

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This paper reviews friction phenomena and friction models of interest for automatic control. Particular emphasis is given to two recently developed dynamic friction models: the Bliman–Sorine model and the LuGre model. These models capture many frictional phenomena observed in laboratory experiments. The behaviours of the models in different situations are discussed in detail. Methods for friction compensation are presented and illustrated with results from practical experiments.

Keywords: Friction; Friction compensation; Models; Observers; Passivity

1. Introduction

Friction occurs in all mechanical systems, e.g. bearings, transmissions, hydraulic and pneumatic cylinders, valves, brakes and wheels. Friction appears at the physical interface between two surfaces in contact. Lubricants such as grease or oil are often used but there may also be a dry contact between the surfaces. Friction is strongly influenced by contaminations. There is a wide range of physical phenomena that cause friction: this includes elastic and plastic deformations, fluid mechanics and wave phenomena, and material sciences [9,10,45,53].

Friction was studied extensively in classical mechanical engineering and there has lately been a strong resurgence. Apart from intellectual curiosity, this is driven by strong engineering needs in a wide range of industries from disc drives to cars. The avail-

ability of new precise measurement techniques has been a good driving force.

Friction is also very important for the control engineer, for example in design of drive systems, high-precision servo mechanisms, robots, pneumatic and hydraulic systems and anti-lock brakes for cars. Friction is highly non-linear and may result in steady state errors, limit cycles, and poor performance. It is therefore important for control engineers to understand friction phenomena and to know how to deal with them. With the computational power available today, it is in many cases possible to deal effectively with friction. This has potential to improve quality, economy and safety of a system.

Friction should be considered early in the system design by reducing it as much as possible through good hardware design. There are, however, cost constraints that may be prohibitive. Dither is a simple way to reduce static friction that has been used for a long time. Dither can be introduced electronically or mechanically by a vibrator, as was done in early auto pilots [41]. Recent advances in computer control have also shown the possibility to reduce the effects of friction by estimation and control. There has also been a significantly increased interest in friction in the control community in terms of special sessions at conferences and in papers.

It is useful for a control engineer to understand the effects of friction on a feedback system and to be able to design control laws that reduce the effects of friction. The purpose of this paper is to review friction models and friction compensation. The new results in the paper are a comparison of two recently developed friction models, the Bliman–Sorine model

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and the LuGre model, and experimental results on friction compensation. A brief description of friction phenomena is given in Section 2. Static models are surveyed in Section 3. Dynamic models are discussed in Section 4; particular attention is given to the Bliman–Sorine model and the LuGre model. The properties of these models are compared in Section 5. Section 6 presents experimental results with friction compensation.

2. Friction Phenomena

Friction is the tangential reaction force between two surfaces in contact. Physically these reaction forces are the results of many different mechanisms, which depend on contact geometry and topology, properties of the bulk and surface materials of the bodies, displacement and relative velocity of the bodies and presence of lubrication.

In dry sliding contacts between flat surfaces, friction can be modelled as elastic and plastic deformation forces of microscopical asperities in contact [9,10]. The asperities each carry a part f_i of the normal load F_N . If we assume plastic deformation of the asperities until the contact area of each junction has grown large enough to carry its part of the normal load, the contact area of each asperity junction is $a_i = f_i/H$, where H is the hardness of the weakest bulk material of the bodies in contact. The total contact area can thus be written $A_R = F_N/H$. This relation holds even with elastic junction area growth, provided that H is adjusted properly. For each asperity contact the tangential deformation is elastic until the applied shear pressure exceeds the shear strength τ_y of the surface materials, when it becomes plastic. In sliding the friction force thus is $F_T = \tau_y A_R$, and the friction coefficient $\mu = F_T/F_N = \tau_y/H$. The friction coefficient does not depend on the normal load or the velocity but it does depend on material properties and contamination.

In dry rolling contact, friction is the result of a non-symmetric pressure distribution in the contact. The pressure distribution is caused by elastic hysteresis in either of the bodies, or local sliding in the contact. For rolling friction, the friction coefficient is proportional to the normal load as $\mu \propto F_N^a$, with $0.2 < a < 1.4$. The elasto-plastic characteristics of dry friction can be described by hysteresis theory [54].

Other physical mechanisms appear when lubri-

cation is added to the contact. At low velocities, the lubricant acts as a surface film, where the shear strength determines the friction. At high velocities and low pressures, a fluid layer of lubricant is built up in the surface. Friction is then determined by shear forces in the fluid layer which depend on the viscosity of the lubricant, and the shear velocity distribution in the fluid film. Approximate expressions for the friction coefficient exist for a number of contact geometries and fluids. At high velocities and pressures, the lubricant layer is built up by elasto-hydrodynamic effects. The lubricant is transformed into an amorphous solid phase due to the high pressure. The shear forces of this solid phase are practically independent of the shear velocity.

The shear strength of a solid lubricant film at low velocities is generally higher than the shear forces of the corresponding fluid film built up at higher velocities. As a result, the friction coefficient in lubricated systems normally decreases when the velocity increases from zero. When the thickness of the film is large enough to completely separate the bodies in contact, the friction coefficient may increase with velocity as hydrodynamic effects become significant. This is called the Stribeck effect.

Film thickness is a vital parameter in lubricated friction. The mechanisms underlying the construction of the fluid film includes dynamics, thus suggesting a dynamic friction model.

Contamination is another factor that adds complication. Small particles of different material between the surfaces give rise to additional forces that strongly depend on the size and material properties of the contaminants.

This short exposé of some friction mechanisms illustrates the difficulty in modelling friction. There are many different mechanisms. It is very difficult to construct a general friction model from physical first principles. Therefore, there have been many attempts to construct phenomenological models that capture the essential phenomena that are relevant for control.

The behaviour of friction has been extensively examined during the 20th century. The experiments have been performed under idealised conditions with clean surfaces and for stationary conditions, e.g. constant velocity. Lately, the interest in friction dynamics has increased. Some experimental observations of friction are reviewed below. The collection is by no means complete but serves to illustrate the many facets of friction behaviour.

2.1. Steady Velocity Friction

The friction force as a function of velocity for constant velocity motion is called the Stribeck curve, after the work of Stribeck [52]. In particular, the dip in the force at low velocities is called the Stribeck effect. The friction-velocity relation is application dependent and varies with material properties, temperature, wear etc. Many friction phenomena do not appear for constant velocity experiments. A number of observations of the dynamic behaviour are given in the following.

2.2. Static Friction and Break-Away Force

Static friction is the friction when sticking. The force required to overcome the static friction and initiate motion is called the break-away force. Many experimental investigations were performed in the 1950s to study the nature of static friction and the break-away force.

Rabinowicz addressed the transition between sticking and sliding [44]. He investigated friction as a function of displacement. He concluded that the break-away force is given by the peak seen in Fig. 1. The maximum friction force typically occurs at a small displacement from the starting point. Johannes et al. [33] found experimentally that the break-away force depends on the rate of increase of the external force. This is confirmed in Richardson and Nolle [48]. A characteristic behaviour is shown in Fig. 2. Another investigation of the behaviour in the sticking regime was done by Courtney-Pratt and Eisner [16]. They studied the spring-like behaviour before gross sliding occurs. Their results were presented in diagrams showing force as a function of displacement (Fig. 3). Note the differences

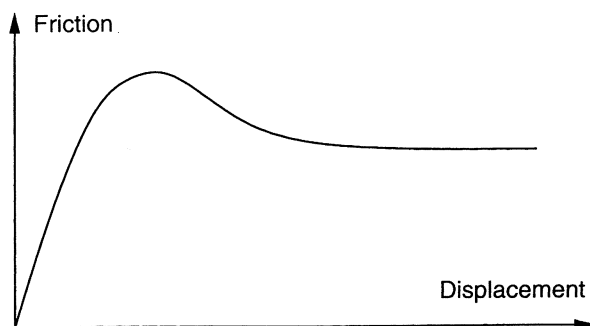


Fig. 1. The relation between friction and displacement as found by Rabinowicz [44]. The experimental results suggested that friction should be described as a function of displacement and not velocity.

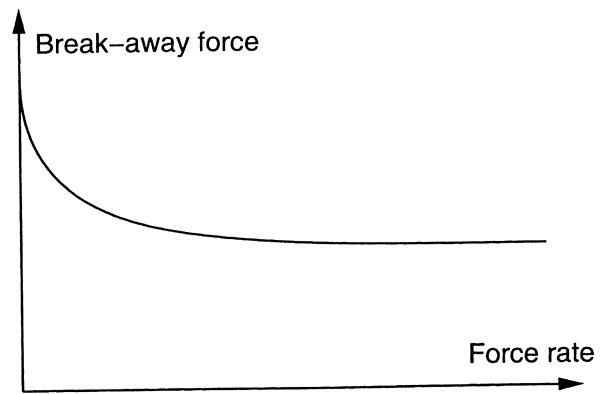


Fig. 2. Characteristic relation between rate of force application and breakaway force as found in Johannes et al. [33]. The experiment suggested that the break-away force decreases with increased rate of force application.

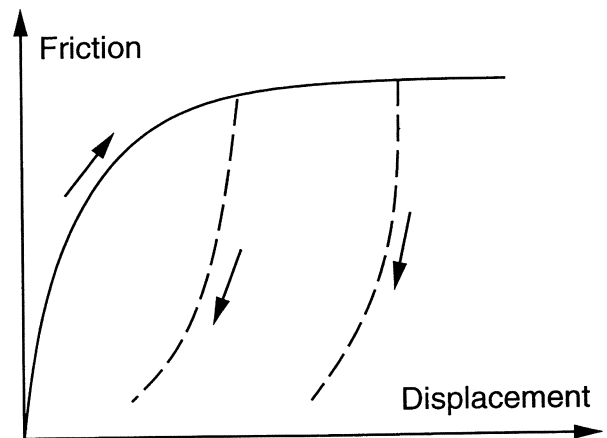


Fig. 3. Pre-sliding displacement as found by Courtney-Pratt and Eisner [16]. The result agrees with Fig. 1 for small displacements. Releasing the applied force results in a permanent displacement as indicated by the dashed lines.

between Figs 1 and 3. The microscopic motion is often called pre-sliding motion.

2.3. Frictional Lag

That dynamics are not only important when sticking was shown by Hess and Soom [32]. They performed experiments with a periodic time-varying velocity superimposed on a bias velocity so that the motion becomes unidirectional. Typically, the friction-velocity relation appeared as in Fig. 4. Hysteresis was observed as the velocity varied. The size of the loop increased with normal load, viscosity and frequency of the velocity variation. These experiments clearly indicate the necessity of using dynamic friction models.

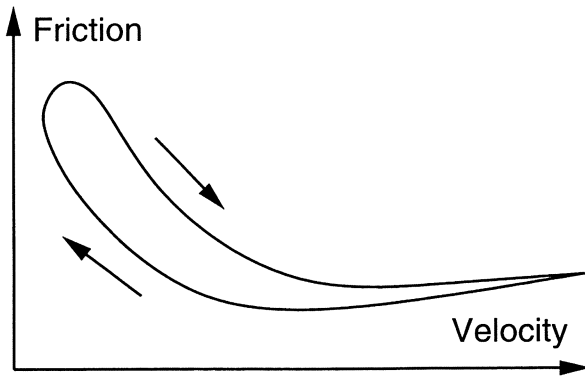


Fig. 4. The friction–velocity relation observed in Hess and Soom [32]. The friction force is lower for decreasing velocities than for increasing velocities. The hysteresis loop becomes wider as the velocity variations become faster.

3. Static Models

In this section, we will give a brief summary of some static friction models.

3.1. Classical Models

The classical models of friction consist of different components, which each take care of certain aspects of the friction force. The main idea is that friction opposes motion and that its magnitude is independent of velocity and contact area. It can therefore be described as

$$F = F_C \operatorname{sgn}(v) \quad (1)$$

where the friction force F_C is proportional to the normal load, i.e. $F_C = \mu F_N$. This description of friction is termed *Coulomb friction*, see Fig. 5(a).

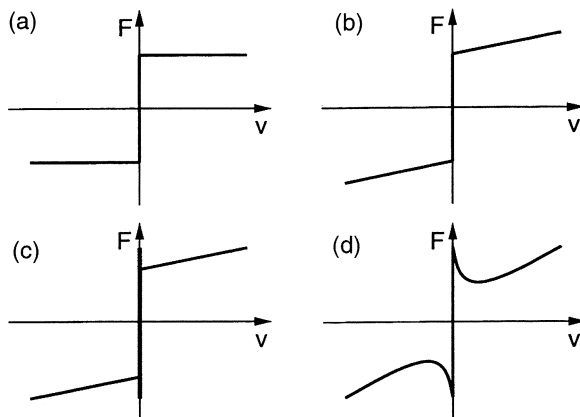


Fig. 5. Examples of static friction models. The friction force is given by a static function except possibly for zero velocity. (a) Coulomb friction; (b) Coulomb plus viscous friction. Stiction plus Coulomb and viscous friction is shown in (c); and (d) shows how the friction force may decrease continuously from the static friction level.

Notice that the model (1) is an ideal relay model. The Coulomb friction model does not specify the friction force for zero velocity. It may be zero or it can take on any value in the interval between $-F_C$ and F_C , depending on how the sign function is defined. The Coulomb friction model has, because of its simplicity, often been used for friction compensation [4,24].

In the 19th century, the theory of hydrodynamics was developed, leading to expressions for the friction force caused by the viscosity of lubricants [47]. The term *viscous friction* is used for this force component, which is normally described as

$$F = F_v v \quad (2)$$

Viscous friction is often combined with Coulomb friction as shown in Fig. 5(b). Better fit to experimental data can often be obtained by a non-linear dependence on velocity. e.g.

$$F = F_v |v|^{\delta_v} \operatorname{sgn}(v) \quad (3)$$

where δ_v depends on the geometry of the application [1,50].

Stiction is short for static friction as opposed to dynamic friction. It describes the friction force at rest. Morin [37] introduced the idea of a friction force at rest that is higher than the Coulomb friction level. Static friction counteracts external forces below a certain level and thus keeps an object from moving.

The friction force for zero velocity is a function of the external force and not the velocity. This can be modelled by

$$F = \begin{cases} F_e & \text{if } v = 0 \text{ and } |F_e| < F_S \\ F_S \operatorname{sgn}(F_e) & \text{if } v = 0 \text{ and } |F_e| \geq F_S \end{cases} \quad (4)$$

The traditional way of depicting friction in block diagrams with velocity as the input and force as the output is therefore not completely correct. If doing so, stiction must be expressed as a multi-valued function that can take on any value between the two extremes $-F_S$ and F_S . Specifying stiction in this way leads to non-uniqueness of the solutions to the equations of motion for the system [8].

The classical friction components can be combined in different ways, as is illustrated in Fig 5(c), and any such combination is referred to as a classical model. These models have components that are affine in velocity. Stribeck [52] observed that the friction force does not decrease discontinuously as in Fig. 5(c), but that the velocity dependence is continuous as shown in Fig. 5(d). This is called *Stribeck friction*. The phenomenon can be described by the model

$$F = \begin{cases} F(v) & \text{if } v \neq 0 \\ F_e & \text{if } v = 0 \text{ and } |F_e| < F_s \\ F_s \operatorname{sgn}(F_e) & \text{otherwise} \end{cases} \quad (5)$$

where $F(v)$ is an arbitrary function, which may look as in Fig. 5(d). A number of parametrisations of $F(v)$ have been proposed [2]. A common form of the non-linearity is

$$F(v) = F_C + (F_S - F_C)e^{-|v/v_\sigma|^{\delta_\sigma}} + F_v v \quad (6)$$

where v_σ is called the Stribeck velocity. Such models have been used for a long time. The function F is easily obtained by measuring the friction force for motions with constant velocity. The curve is often asymmetrical.

3.2. The Karnopp Model

The main disadvantage when using a model such as (5), for simulation or control, is to determine when the velocity is zero. A remedy for this is found in the model presented by Karnopp [34]. It was developed to overcome the problems with zero velocity detection and to avoid switching between different state equations for sticking and sliding. The model defines a zero velocity interval, $|v| < DV$. For velocities within this interval the internal state of the system (the velocity) may change and be non-zero but the output of the block is maintained at zero by a dead-zone. Depending on if $|v| < DV$ or not, the friction force is either a saturated version of the external force or an arbitrary static function of velocity. The interval $\pm DV$ can be quite coarse and still promote so called stick-slip behaviour.

The drawback with the model is that it is strongly coupled to the rest of the system. The external force is an input to the model and this force is not always given explicitly. The model therefore has to be tailored for each configuration. Variations of the Karnopp model are widely used since they allow efficient simulations. The zero velocity interval, however, does not agree with real friction.

3.3. Armstrong's Model

To account for some of the observed dynamic friction phenomena, a classical model can be modified as proposed by Armstrong-Helouvry et al. [3]. This model introduces temporal dependencies for stiction and the Stribeck effect, but does not handle pre-sliding displacement. This is instead done by describing the sticking behaviour by a separate equa-

tion. Some mechanism must then govern the switching between the model for sticking and the model for sliding. The friction is described by

$$F(x) = \sigma_0 x \quad (7)$$

when sticking and by

$$F(v, t) = \left(F_C + F_S(\gamma, t_d) \frac{1}{1 + (v(t - \tau_e)/v_S)^2} \right) \operatorname{sgn}(v) + F_v v \quad (8)$$

when sliding, where

$$F_S(\gamma, t_d) = F_{S,a} + \left(F_{S,\infty} - F_{S,a} \frac{t_d}{t_d + \gamma} \right) \quad (9)$$

$F_{S,a}$ is the Stribeck friction at the end of the previous sliding period and t_d the dwell time, i.e., the time since becoming stuck. The sliding friction (8) is equivalent to a static model where the momentary value of the velocity in the Stribeck friction has been replaced by a delayed version and where it has a time-dependent coefficient. The model requires seven parameters.

Since the model consists of two separate models, one for sticking and one for sliding, a logical statement, probably requiring an eighth parameter, determines the switching. Furthermore, the model states have to be initialised appropriately every time a switch occurs.

4. Dynamic Models

Lately, there has been a significant interest in dynamic friction models. This has been driven by intellectual curiosity, demands for precision servos and advances in hardware that make it possible to implement friction compensation. In this section, we will summarise several dynamic models.

4.1. The Dahl Model

The Dahl model [18] was developed for the purpose of simulating control systems with friction. The model is discussed elsewhere [19,21]. It has been used for adaptive friction compensation [23,53]. Dahl's began with several experiments on friction in servo systems with ball bearings. One of his findings was that bearing friction and solid friction have similar behaviour. Dahl's experiments indicate that there are metal contacts between the surfaces. Dahl developed a comparatively simple model that

was used extensively to simulate systems with ball bearing friction.

The starting point for Dahl's model is the stress-strain curve in classical solid mechanics [46,49] (Fig. 6). When subject to stress, the friction force increases gradually until rupture occurs. Dahl modelled the stress-strain curve by a differential equation. Let x be the displacement, F the friction force, and F_c the Coulomb friction force. Then Dahl's model has the form

$$\frac{dF}{dx} = \sigma \left(1 - \frac{F}{F_c} \operatorname{sgn} v \right)^\alpha$$

where σ is the stiffness coefficient and α is a parameter that determines the shape of the stress-strain curve. The value $\alpha = 1$ is most commonly used. Higher values will give a stress-strain curve with a sharper bend. The friction force $|F|$ will never be larger than F_c if its initial value is such that $|F(0)| < F_c$.

Notice that in this model the friction force is only a function of the displacement and the sign of the velocity. This implies that the friction force is only position dependent. This so-called rate independence is an important property of the model. It makes it possible to use the theory of hysteresis operators [35]. It is also used in extensions of the model [5].

To obtain a time domain model, Dahl observed that

$$\begin{aligned} \frac{dF}{dt} &= \frac{dF}{dx} \frac{dx}{dt} = \frac{dF}{dx} v \\ &= \sigma \left(1 - \frac{F}{F_c} \operatorname{sgn} v \right)^\alpha v \end{aligned} \quad (10)$$

The model is a generalisation of ordinary Coulomb friction. The Dahl model neither captures the Striebeck effect, which is a rate-dependent phenomenon,

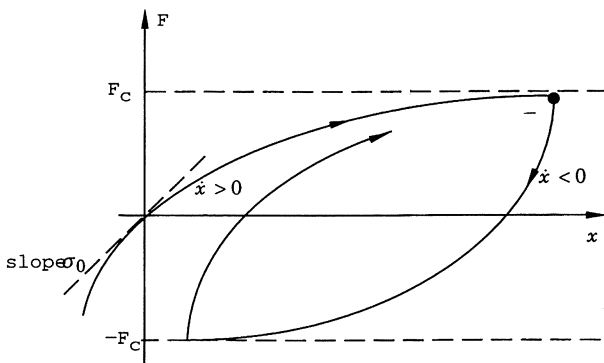


Fig. 6. Friction force as a function of displacement for Dahl's model.

nor does it capture stiction. These are the main motivations for the recent extensions of the model [5,13].

For the case $\alpha = 1$ the Dahl model (10) becomes

$$\frac{dF}{dt} = \sigma v - \frac{F}{F_c} |v|$$

Introducing $F = \sigma z$ the model can be written as

$$\frac{dz}{dt} = v - \frac{\sigma |v|}{F_c} z \quad (11)$$

$$F = \sigma z$$

4.2. The Bristle Model

Haessig and Friedland [28] introduced a friction model which attempted to capture the behaviour of the microscopical contact points between two surfaces. Due to irregularities in the surfaces, the number of contact points and their location are random. Each point of contact is modelled as a bond between flexible bristles. As the surfaces move relative to each other, the strain in the bond increases and the bristles act as springs. The force is then given by

$$F = \sum_{i=1}^N \sigma_0 (x_i - b_i) \quad (12)$$

where N is the number of bristles, σ_0 the stiffness of the bristles, x_i the relative position of the bristles, and b_i the location where the bond was formed. As $|x_i - b_i|$ equals δ_s the bond snaps and a new one is formed at a random location relative to the previous location.

The complexity of the model increases with N . Good results were found with 20–25 bristles, but even a single bristle gave reasonable qualitative behaviour. The stiffness of the bristles, σ_0 , can be made velocity dependent. An interesting property of the model is that it captures the random nature of friction. The randomness depends on the number of bristles. The model is inefficient in simulations due to its complexity. Motion in sticking may be oscillatory since there is no damping of the bristles in the model.

4.3. The Reset Integrator Model

Haessig and Friedland [28] also proposed the reset integrator model. This model can be viewed as an attempt to make the bristle model computationally feasible. Instead of snapping a bristle, the bond is

kept constant, limiting the increase of the strain at the point of rupture. The model utilises an extra state to determine the strain in the bond, which is modelled by

$$\frac{dz}{dt} = \begin{cases} 0 & \text{if } (v > 0 \text{ and } z \geq z_0) \text{ or } (v < 0 \text{ and } z \leq -z_0) \\ v & \text{otherwise} \end{cases}$$

The friction force is given by

$$F = (1 + a(z))\sigma_0(v)z + \sigma_1 \frac{dz}{dt} \quad (13)$$

where $\sigma_1 dz/dt$ is a damping term that is active only when sticking. The damping coefficient can be chosen to give a desired relative damping of the resulting spring–mass–damper system. Stiction is achieved by the function $a(z)$, which is given by

$$a(z) = \begin{cases} a & \text{if } |z| < z_0 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

If $|z| < z_0$ the model describes sticking where the friction force is a function of z . As the deflection reaches its maximum value z_0 , the variable z remains constant and the friction force drops since $a(z)$ becomes zero. The friction force when slipping is an arbitrary function of the velocity given by $\sigma_0(v)$. The reset integrator model is much simpler to simulate than the bristle model, but some care has to be used to deal with the discontinuities.

4.4. The Models by Bliman and Sorine

Bliman and Sorine have developed a family of dynamic models in a series of papers [6–8]. It is based on the experimental investigations by Rabinowicz [44].

Bliman and Sorine stress rate independence. The magnitude of the friction depends only on $\text{sgn } v$ and the space variable s defined by

$$s = \int_0^t |v(\tau)| d\tau.$$

In the Bliman–Sorine models, friction is then a function of the path only. It does not depend on how fast the system moves along the path. This makes it possible to use the elegant theory of hysteresis operators developed elsewhere [35,54]. The models are expressed as linear systems in the space variable s .

$$\frac{dx_s}{ds} = Ax_s + Bv_s \quad (15)$$

$$F = Cx_s$$

The variable $v_s = \text{sgn}(v)$ is required to obtain the correct sign. Bliman and Sorine have models of different complexity. The first-order model is given by

$$A = -1/\epsilon_f, B = f_1\epsilon_f \text{ and } C = 1 \quad (16)$$

This model can be written as

$$\frac{dF}{dt} = \frac{dF}{ds} \frac{ds}{dt} = |v| \frac{dF}{ds} = f_1/\epsilon_f \left(v - |v| \frac{F}{f_1} \right)$$

which is identical to the Dahl model (10) with $F_C = f_1$, $\sigma = f_1/\epsilon_f$ and $\alpha = 1$. The first-order model does not give stiction, nor does it give a friction peak at a specific break-away distance as observed by Rabinowicz. This can, however, be achieved by a second-order model with

$$A = \begin{pmatrix} -1/(\eta\epsilon_f) & 0 \\ 0 & -1/\epsilon_f \end{pmatrix} \quad (17)$$

$$B = \begin{pmatrix} f_1/(\eta\epsilon_f) \\ -f_2/\epsilon_f \end{pmatrix} \text{ and } C = \begin{pmatrix} 1 & 1 \end{pmatrix}$$

where $f_1 - f_2$ corresponds to kinetic friction reached exponentially as $s \rightarrow \infty$ [8]. The model (17) can be viewed as a parallel connection of a fast and a slow Dahl model. The fast model has higher steady state friction than the slow model. The force from the slow model is subtracted from the fast model, which results in a stiction peak. Both the first- and second-order models can be shown to be dissipative. Bliman and Sorine also show that, as ϵ_f goes to zero, the first-order model behaves as a classical Coulomb friction model, and the second-order model as a classical model with Coulomb friction and stiction. It should be noted that the Stribeck effect of the second-order model, claimed by the authors, is not the same as observed in Stribeck [52]. The emulated effect by the second-order model is only present at a certain distance after motion starts. This means that it will not appear when the motion slows down, as the true Stribeck effect would. The friction peak is instead the equivalent of stiction for a dynamic model.

4.5. Models for Lubricated Contacts

The friction interfaces in most engineering applications are lubricated. Friction models have therefore been derived using hydrodynamics. Viscous friction is a simple example, but other models also exist. In

Harnoy and Friedland [30], a model based on the hydrodynamics of a lubricated journal bearing is introduced. The model stresses the dynamics of the friction force. The eccentricity ϵ of the bearing is an important variable in determining the friction force. A simplified model is given by

$$F = K_1(\epsilon - \epsilon_{tr})^2 \Delta + \frac{K_2}{\sqrt{1 - \epsilon^2}} v$$

The first term is due to the shearing of the asperity contacts and the second term is due to the viscosity of the lubricant. The function Δ is an indicator function that is one for $\epsilon > \epsilon_{tr}$ and zero otherwise. This implies that, for small eccentricities, there is no friction due to asperity contacts. The eccentricity is given by a fourth-order differential equation, which determines the pressure distribution in the lubricant. The model requires five parameters. Simulations show a behaviour very similar to the observations in Hess and Soom [32]. An extension including sleeve compliance is given in Harnoy and Friedland [31]. The model then becomes even more complicated and requires determination of initial values when switching between slipping and sticking.

4.6. The LuGre Model

The LuGre model is a dynamic friction model presented in Canudas de Wit et al. [13]. Extensive analysis of the model and its application can be found in Olsson [39]. The model is related to the bristle interpretation of friction as in Haessig and Friedland [28]. Friction is modelled as the average deflection force of elastic springs. When a tangential force is applied, the bristles will deflect like springs. If the deflection is sufficiently large, the bristles start to slip. The average bristle deflection for a steady state motion is determined by the velocity. It is lower at low velocities, which implies that the steady state deflection decreases with increasing velocity. This models the phenomenon that the surfaces are pushed apart by the lubricant, and models the Stribeck effect. The model also includes rate-dependent friction phenomena such as varying break-away force and frictional lag. The model has the form

$$\begin{aligned} \frac{dz}{dt} &= v - \sigma_0 \frac{|v|}{g(v)} z \\ F &= \sigma_0 z + \sigma_1(v) \frac{dz}{dt} + f(v) \end{aligned} \quad (18)$$

where z denotes the average bristle deflection. The model behaves like a spring for small displacements. Linearisation of (18) around zero velocity and zero state gives

$$\begin{aligned} \frac{d(\delta z)}{dt} &= \delta v, \\ \delta F &= \sigma_0 \delta z + (\sigma_1(0) + f'(0)) \delta v \end{aligned}$$

The parameter σ_0 is the stiffness of the bristles, and $\sigma_1(v)$ the damping. For constant velocity, the steady state friction force is

$$F = g(v) \operatorname{sgn}(v) + f(v) \quad (19)$$

The function $g(v)$ models the Stribeck effect, and $f(v)$ is the viscous friction. A reasonable choice of $g(v)$ which gives a good approximation of the Stribeck effect is

$$g(v) = \alpha_0 + \alpha_1 e^{-(v/v_0)^2} \quad (20)$$

compare with (6). The sum $\alpha_0 + \alpha_1$ then corresponds to stiction force and α_0 to Coulomb friction force. The parameter v_0 determines how $g(v)$ vary within its bounds $\alpha_0 < g(v) \leq \alpha_0 + \alpha_1$. A common choice of $f(v)$ is linear viscous friction $f(v) = \alpha_2 v$ as in (2), see also (3).

The following special case of the model given by Eqs (18) and (20), which has linear viscous friction and constant σ_1 , is called the *standard parametrization*.

$$\begin{aligned} \frac{dz}{dt} &= v - \sigma_0 \frac{|v|}{g(v)} z \\ g(v) &= \alpha_0 + \alpha_1 e^{-(v/v_0)^2} \\ F &= \sigma_0 z + \sigma_1 z + \alpha_2 v \end{aligned} \quad (21)$$

It is useful to let the damping σ_1 decrease with increasing velocity, e.g.

$$\sigma_1(v) = \sigma_1 e^{-(v/v_d)^2} \quad (22)$$

Physically this is motivated by the change of the damping characteristics as velocity increases, due to more lubricant being forced into the interface. Another reason for using (22) is that it gives a model which is dissipative [39].

5. Comparison of the Bliman–Sorine and the LuGre Models

The Bliman–Sorine (15) and the LuGre models (21) are both extensions of the Dahl model (10). The Dahl model (10) has many attractive features. It is

a dynamic model that captures many aspects of friction. The model is so simple that it can be used for model-based friction compensation. It has, however, a serious drawback because it does not describe stiction. The Bliman–Sorine and the LuGre models both capture stiction and the Stribeck effect. Bliman and Sorine use two Dahl models in parallel. The LuGre model is a first-order model with a coefficient that depends on velocity. The models have many similarities but also significant differences.

5.1. Rate Dependency

The LuGre model is inherently rate dependent. The Bliman–Sorine model is seemingly independent of rate because it is expressed in terms of the space variable s . There are, however, some subtleties because the term $v_s = \text{sgn } v(t)$ enters the right-hand side of Eq. (15). The variable v_s which takes the value 1 or -1 changes sign when the velocity changes sign. This introduces a special kind of time dependency in the model, because there will be a transient when the velocity changes sign. After the transient the friction will settle to the steady state solution of Eq. (15), which is given by

$$F^0 = -CA^{-1}Bv_s \quad (23)$$

The transient which makes F different from F^0 is the mechanism that gives rise to stiction. The shape of the transient, and therefore the friction characteristics of the model, is dependent on whether the velocity reversal takes place from a steady state or not. As a result, the transient will cause difficulties when there are rapid changes of sign of the velocity.

The ideal model trajectory going from steady state for $-v_s$ to steady state for v_s without sign reversal in velocity is written

$$F^*(s, v_s) = Ce^{As} \left(A^{-1} + \int_0^s e^{-A\sigma} d\sigma \right) Bv_s \quad (24)$$

This trajectory is also defined by the unique solution $x_s(0) = -F^0$ of

$$\max_{x_s(0), s} F(s) \quad (25)$$

for constant v_s , i.e. (24) is the trajectory giving the largest break-away friction force. Consequently, trajectories that do not start from steady state for $s = 0$ give different friction characteristics than F^* . For example, model trajectories starting with zero states give a stiction peak $f_s - f_k$ only half that of $F^*(s)$.

5.2. Oscillatory Behaviour at Low Velocities

A simple experiment that reveals much about friction is to explore the open loop behaviour of a drive system. Let J be the moment of inertia, F the friction torque, and u an external driving torque. The system is described by

$$J \frac{d^2x}{dt^2} + F = u \quad (26)$$

Figures 7 and 8 show the responses to a sinusoidal input torque $u = 0.4 \sin(t)$ [Nm] for the Bliman–Sorine and the LuGre models.

With the Bliman–Sorine model shown in Fig. 7 there are significant oscillations in the friction force in the stick mode. Because of the oscillations, the friction force F differs from F^* given by Eq. (24). One effect of this is that the break-away force becomes difficult to predict. As noted above, the zero initial conditions used in the simulation result in a smaller first stiction peak, resulting in a remaining bias in position. In degenerate cases with non-steady state initial conditions and oscillatory behaviour a symmetric periodic input force can give a unidirectional stick-slip motion.

With the LuGre model shown in Fig. 8 there are rapid changes in the friction force which brings the system quickly to rest. Compare Canudas de Wit et al. [14] for a detailed discussion. There are, however, no oscillations in the friction force and no remaining bias in the displacement. All stiction peaks are of the same magnitude.

The oscillatory behaviour of the Bliman–Sorine model is further illustrated in Fig. 9, which shows the phase plane. The friction torque $F = x_1 + x_2$ is shown in the figure. We also show the trajectory that would be obtained with the Friction F^* . Note that the oscillations force the trajectory of F inside the trajectory of F^* , resulting in a lower friction torque in the transient. The behaviour shown in Fig. 7 is clearly not desirable.

5.3. Damping

Insight into the differences between the models illustrated in Figs 7 and 8 can also be obtained by linearising the model (26) for motions with small velocity. We get

$$\frac{d^2x}{dt^2} + 2\zeta\omega \frac{dx}{dt} + \omega^2\theta = \frac{u}{J} \quad (27)$$

The Bliman–Sorine model with linear viscous friction has the relative damping

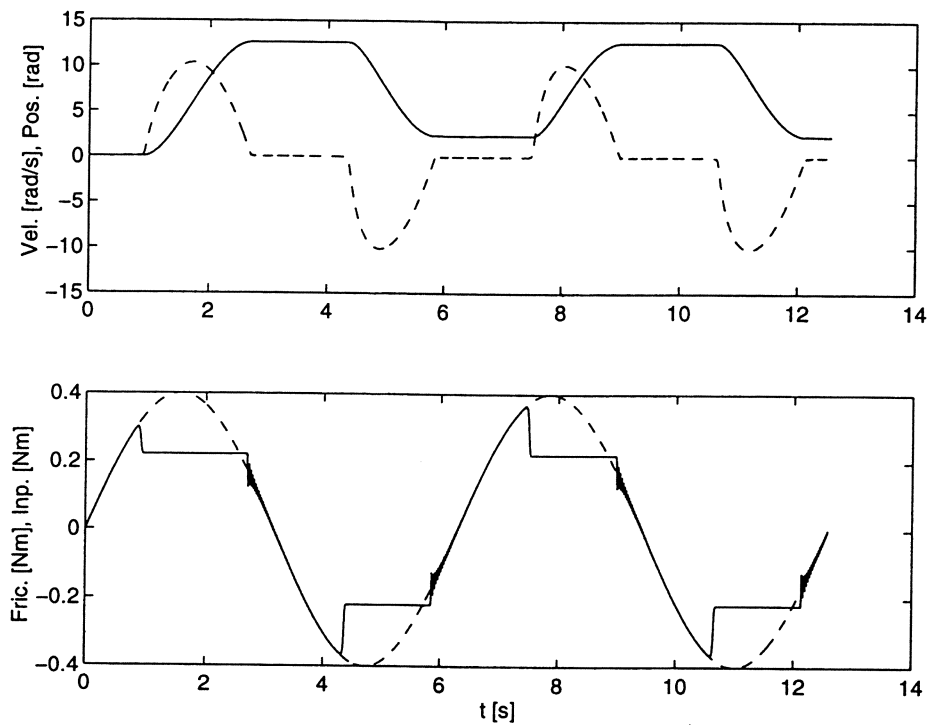


Fig. 7. Response of the system (26) with Bliman-Sorine friction to a small sinusoidal disturbance torque.

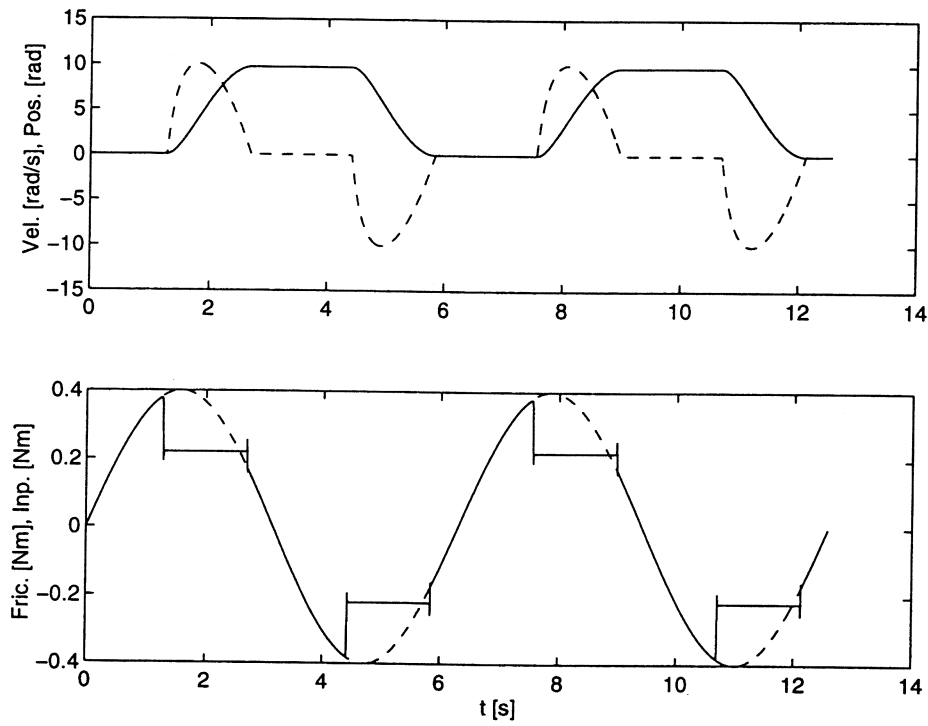


Fig. 8. Response of the system (26) with LuGre friction to a small sinusoidal disturbance torque.

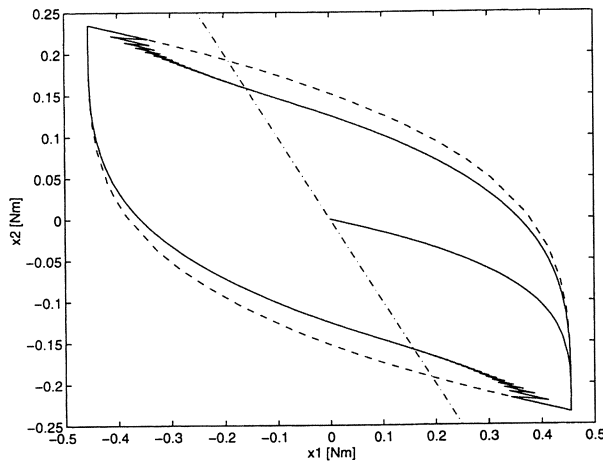


Fig. 9. Phase plane corresponding to Fig. 7 for the Bliman–Sorine model. The ideal trajectory F^* is also shown in dashed lines. The dashed-dotted line is the curve where the friction is zero.

$$\zeta = \frac{F_v}{2} \sqrt{\frac{\epsilon_f \eta}{J f_1 - \eta f_2}} \quad (28)$$

The damping is thus proportional to the viscous friction coefficient F_v . This means that a low velocity property for the stick region is determined entirely by a high velocity property in slip, and cannot be chosen freely. The relative damping for the LuGre model is

$$\zeta = \frac{\sigma_1 + \alpha_2}{2\sqrt{J\sigma_0}} \quad (29)$$

The damping thus depends on the parameter σ_1 which can be chosen freely independent of the viscous friction coefficient α_2 .

5.4. Dissipativity

Dissipativity is a very desirable property of a friction model. The Bliman–Sorine model is dissipative. The LuGre model is dissipative if

$$\sigma_1 < 4 \frac{g(v)}{|v|} \quad (30)$$

A possible velocity dependence is given by Eq. (22).

5.5. Behaviour for Small Displacements

The behaviour for small displacements is of particular interest for control, particularly in applications that involve precision pointing or positioning. For small displacements, the system operates in a region where the friction force changes very rapidly.

An experiment that reveals much about the behaviour at small displacements is to apply an input force $T = b + \alpha \sin \omega t$ to the system described by Eq. (26). It is interesting to have a close to b so that the velocity will be small for a long period. This can be enhanced by choosing waveforms where the force is close to zero for even longer periods. It is interesting to separate the case when $b \geq a$, which implies that the force is unidirectional, from $b < a$ when there may be velocity reversals. The behaviour is very different in these cases. The behaviour obtained with the different friction models is also quite different.

Early experiments of this type are described in Courtney-Pratt and Eisner [17]. Similar experiments with servo drives were described in Canudas de Wit et al. [11]. The data in this section are from Gäfvert [25]. They were obtained from experiments with a DC servo with a gearbox. The servo drive is well lubricated. A mechanical brake was used to increase dry friction. It is easy to perform experiments with a unidirectional torque, i.e. $b \geq a$. Because of the gearbox, it is difficult to make meaningful experiments with a torque that changes sign because the backlash in the gearbox may hide the friction effects.

Figure 10 shows the responses. The behaviours are similar for small input torques, when typical hysteresis motion is exhibited. The behaviour changes with increasing torque level. For larger input torques, the behaviour is elastic when dry friction dominates and plastic in the lubricated case. In both cases, the elastic stiffness changes rapidly at certain displacements. The Bliman–Sorine and the LuGre models do not separate between the cases when dry friction or viscous friction dominates. These cases only differ in terms of the values of the parameters. Simulations of the experiment were carried out for both models. The results are shown in Fig. 11. The friction models' parameters are the result of identification on the servo. The LuGre identification procedure results in model parameters adapted to the large initial stiffness of the servo, whereas the Bliman–Sorine model identification gives parameters adapted to the overall pre-sliding stiffness. As a consequence, the LuGre model underestimates the displacements in the simulations. The model responses differ significantly from the experimental results. This behaviour will not change substantially if the parameters are changed. The models do not succeed in reproducing the behaviour with a hysteresis loop with slip found in the experiment. See Ganseman [26] for an extension of the LuGre model to approach to this problem, and Dahl and Wilder [20] for a correction of the Dahl model.

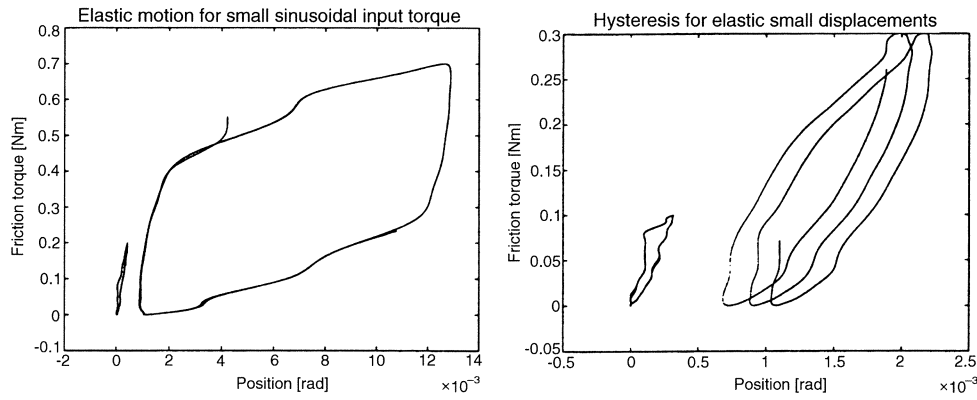


Fig. 10. Results of experiments with a sinusoidal torque $T = b + a \sin t$. Dry friction dominates in the curves on the left where a brake is applied. The parameters are $a = b = 0.05$ Nm for the small hysteresis loop and $a = b = 0.35$ for the large loop. Lubricated friction dominates in the figure on the right where the parameters are $a = b = 0.10$ Nm for the small hysteresis loop and $a = b = 0.15$ Nm for the large loop.

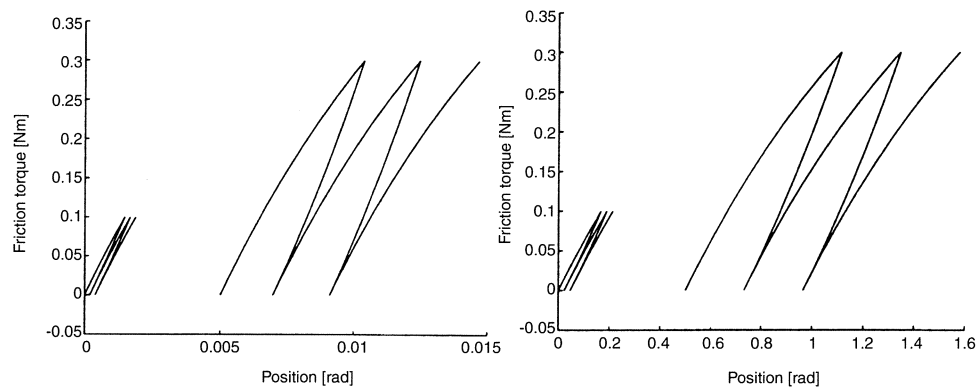


Fig. 11. Simulation of the experiment in Fig. 10 with the Bliman–Sorine model (left) and the LuGre model (right). The parameters are $a = b = 0.05$ Nm for the small hysteresis curve and $a = b = 0.15$ Nm for the large hysteresis curve.

Notice, however, that both models give the correct qualitative behaviour for experiments where the perturbations' torque changes sign. This is illustrated for the LuGre model in Fig. 12, which shows simu-

lations for the case $b < a$. Notice that closed hysteresis loops occur in both cases and that there is slip in the case $b \neq 0$. This observation may provide useful hints for modification of the model.

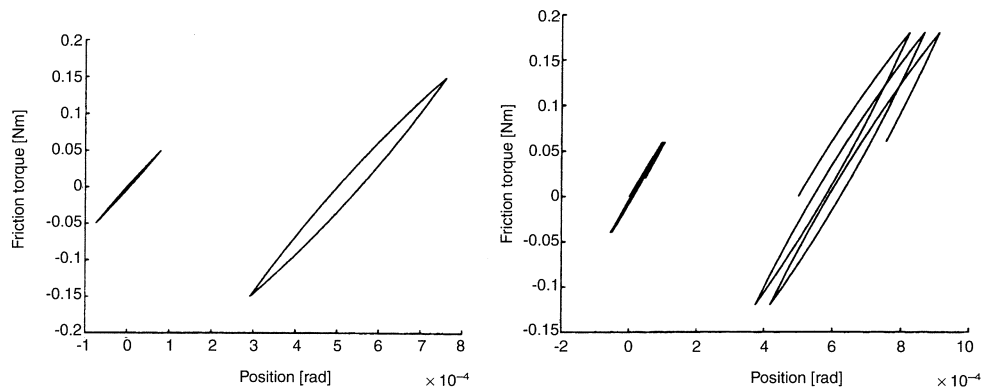


Fig. 12. Simulation of experiments with perturbation torques that change sign with the LuGre model. In the left figure, the parameters are $a = 0.05$ Nm, $b = 0$ Nm for the small hysteresis loop and $a = 0.15$ Nm, $b = 0$ Nm for the large hysteresis loop. In the right figure, the parameters are $a = 0.05$ Nm, $b = 0.01$ Nm for the small hysteresis loop and $a = 0.15$ Nm, $b = 0.3$ Nm for the large hysteresis loop.

6. Control Systems Applications

In control engineering, it is of interest to model systems with friction to better understand their behaviour and to design control strategies that can alleviate the performance deterioration due to friction. This is of interest in simple devices such as standard process control loops with valves and in complicated precision systems for accurate pointing, for example telescopes, radar, robots and gun turrets.

6.1. Friction Compensation

There are many ways to compensate for friction. A simple way to eliminate some effects of friction is to use a dither signal, that is, a high-frequency signal that is added to the control signal. An interesting form of this was used in gyroscopes for autopilots in the 1940s. The dither signal was generated by a mechanical vibrator [41]. The dither signal eliminates stiction by keeping the system in continuous motion. A modern version of dither is the Knocker, introduced in Hägglund [29], for use in industrial valves. The effects of dither in systems with dynamic friction (LuGre) was recently studied [43].

Systems for motion control typically have a cascade structure with a current loop, a velocity loop and a position loop. Since friction appears in the inner loop it would be advantageous to introduce friction compensation in that loop. This is difficult to do with conventional systems because it is not easy to modify the current loop. Because of the price and performance of micro-electronics there is a trend that current loops are implemented with computer control. With such implementations, it is natural to make the friction compensation in the inner loop.

To obtain an effective friction compensation it is necessary that the velocity is measured or estimated with good resolution and small time delay. Friction compensation is more difficult if there is considerable dynamics between the control signal and the friction force. The sensor problem can be considerable with a shaft encoder because there will be a variable delay in estimation of the velocity.

If a good friction model is available it is possible to use a model-based friction compensation scheme. The idea is very simple. The friction force F is estimated using some model, and a signal that compensates the estimated friction force $\hat{F}$ is added to the control signal. This is illustrated in the block diagram in Fig. 13.

For tracking tasks, friction can be predicted and partially compensated by feedforward. This has the advantage of eliminating the lag and the noise effects of the velocity prediction. It is only suitable for tracking since the desired velocity trajectory is known in advance.

An early application to an optical telescopic drive is presented in Gilbert and Winston [27]. A model reference adaptive system was designed based on Coulomb friction. Canudas de Wit et al. [11] describes an application to velocity control. A static friction model with Coulomb and viscous friction is used. The model allows for asymmetric friction characteristics. The measurement of the velocity is critical. It is also useful not to over-compensate the friction, since this may lead to instabilities. The system has been used extensively in the control laboratory. It requires occasional adjustments of the friction parameters and gives good results. Schemes of this type are now standard in motion control systems. The performance of the friction compensation can be improved by using more elaborate friction models and by adapting their parameters.

Friction compensation for a position servo with velocity and position control based on the LuGre model is described in Canudas de Wit et al. [13]. The design is based on passivity. It is shown that the system can be decomposed into a standard feedback configuration with a linear block and a non-linear block. Passivity theory is used to derive conditions on the controller that guarantees that the closed loop system is stable. The condition is that the resulting linear block is strictly positive real (SPR). In Olsson [39] and Olsson and Åström [40] it is shown that it is sufficient to have a linear block that is positive real (PR). This is important from a practical point of view because it applies to the case where there is integral action in the inner loop. The LuGre model has also been used in connection with more complex non-linear systems, such as robot manipulators. In this case, the parametrisation [22], which renders the friction model passive for all velocities, can be used explicitly in the design of passivity-based tracking controllers for rigid robots [22]. In this context, passivity is used to generalise the SPR condition of the upper-block linear operator in Canudas de Wit et al. [13].

Another approach to friction compensation using the LuGre model is presented in Panteley et al. [42]. A strategy for global tracking of a robot manipulator is presented. Measurements of position and velocity are available and all system parameters for the robot and the friction model are unknown. This is achieved with an adaptive controller that dominates the

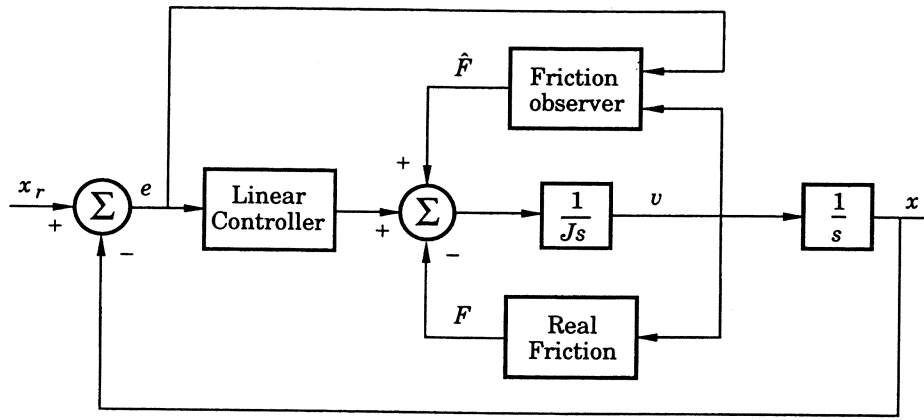


Fig. 13. Block diagram of the observer-based friction compensation scheme.

effects of friction (in a suitable Lyapunov function derivative) with a function of the measured signals. The controller is non-smooth, but the stabilisation does not rely on the generation of sliding regimes.

The Bliman and Sorine model can also be used for friction compensation. Control design is done directly in the spatial coordinates instead of in the time domain. The first-order model [16] (which coincides with the Dahl model) has been used for friction compensation to improve the accuracy on optronic systems [51]. Applications of the Bliman–Sorine models are particularly suitable for systems with dominant elasto-plastic effects.

6.2. Velocity Tracking

Friction compensation based on the LuGre model for a servo mechanism is described in Olsson and Åström [38]. The compensation scheme is based on an observer giving an estimate of the friction. The system regarded is described by

$$\frac{dv}{dt} = u - F$$

$$y = v + n$$

where v is velocity, F friction and n a velocity disturbance given as band-limited Gaussian noise. The control objective is to keep y close to zero in the presence of the disturbance n . The realisation of the disturbance used in the simulations is shown in Fig. 14. A linear PI controller is designed based

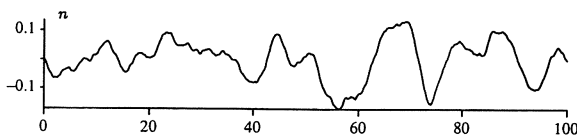


Fig. 14. Velocity disturbance for the servo problem.

on the linear dynamics, and a model-based friction compensation is added. The control signal thus is composed of a linear and a non-linear part $u = u_{lin} + \hat{F}$, where u_{lin} is the linear part and $\hat{F}$ is the estimated friction force. Figure 15 shows a simulation of the system with a linear PI controller without friction compensation. The figure shows the tracking error, the friction torque and the linear control signal. Notice that there are large errors and large jumps in the control signal when the velocity passes through zero. The jumps are caused by friction. An analysis of the control error shows that its amplitude distribution is far from Gaussian: it has heavy tails. Also notice the rapid rise in friction force which occurs when the velocity approaches zero.

The performance improvements obtained with friction compensation based on the LuGre model are shown in Fig. 16. The maximum tracking error is reduced by an order of magnitude. Analysis of the tracking error shows that the amplitude distribution is close to Gaussian. This observation indi-

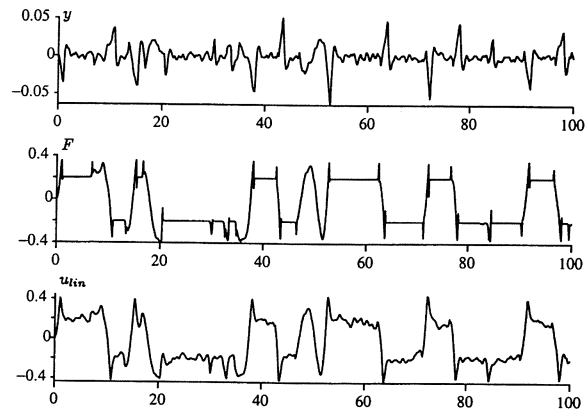


Fig. 15. Illustration of performance degradation due to friction in a servo system.

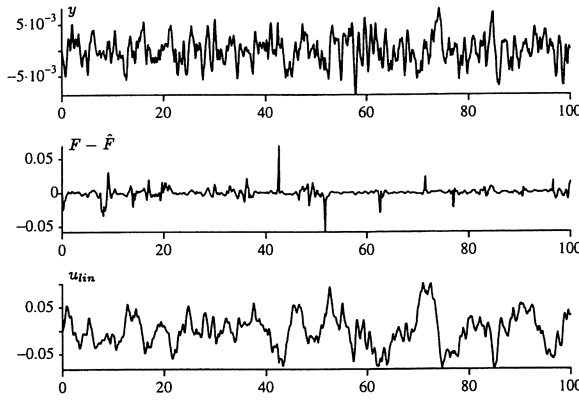


Fig. 16. Behaviour of the tracking system in Fig. 15 when friction compensation based on the LuGre model is used.

cates that the presence of friction can be determined from the amplitude distribution of tracking errors, provided that noise that enters the system is Gaussian. A Gaussian distribution indicates that there is no friction; deviations from the Gaussian distribution indicates that there is friction. A detailed investigation is given in Olsson and Åström [38]. This includes comparisons with other friction models and sensitivity analysis. The measures

$$e_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T e^2(\tau) d\tau}$$

and

$$e_{\text{max}} = \max_{t \in [0, T]} |e(t)|$$

are used to evaluate the results. A summary of some results are given in Table 1.

When there is no friction, the root mean square of the tracking error is $3.12 \cdot 10^{-3}$. This value increases to $13.0 \cdot 10^{-3}$ in the presence of friction. The RMS error is reduced to $7.85 \cdot 10^{-3}$ by friction compensation based on a Coulomb friction model and further to $2.65 \cdot 10^{-3}$ with a compensator based on the LuGre model. Notice that this value is smaller

Table 1. Investigations of the effect of friction on a tracking mechanism, and the impact of some different compensation strategies

	$e_{\text{rms}} \cdot 10^3$	$e_{\text{max}} \cdot 10^3$
No friction	3.12	9.06
With friction	13.0	63.7
Friction compensation (Coulomb)	7.85	32.7
Friction compensation (LuGre)	2.65	8.57
Overcompensation	6.72	28.5
Undercompensation	6.22	25.8

than the value obtained for the system without friction. The reason for this is that a PI controller was used. The dynamics of the friction observer reduces the error. In the table we also show the results when the friction force parameter in the observer is 50% too large (over-compensation) and 50% too small (under-compensation). In these cases the RMS error increases to $6.72 \cdot 10^{-3}$ and $6.22 \cdot 10^{-3}$ respectively. The values indicate that the friction observer is not overly sensitive to parameter variations, but that there is a substantial incentive to use an adaptive scheme for friction compensation. Such a scheme will be described at the end of this section.

6.3. Laboratory Experiments

A test rig for experimenting with control of systems with friction has been built at INPG in Grenoble (Fig. 17). This equipment has a brake which makes it easy to increase friction and it is interfaced to the dSPACE signal processing equipment which makes it easy to do experiments flexible. The equipment has been used extensively for the results discussed in this paper.

The standard LuGre model (21) has been used to design friction compensators. The parameters of the model were determined by system identification experiments described in Canudas de Wit and Lischinsky [15]. The experimental conditions were varied to emphasise different properties of the model. Experiments with ‘large’ displacements were used to estimate the steady state friction characteristics during motion in sliding, and experiments with ‘small’ displacements in the stiction zone were used to determine stiffness and damping. The parameter values varied with operating conditions. Some representative values are given in Table 2.

An alternative for parameter identification is proposed by Bliman and Sorine. Their idea is to determine key features of the hysteresis plot of friction force versus displacements during a cyclic motion with sign changes in velocity, typically maximum overshooting (in amplitude) and setting value (in distance). They have shown that, under certain conditions, there exists a one-to-one map between these points and the four parameters of model (17). Unfortunately, the conditions do not hold when the friction torque drops too fast after the break-away. This is the case for the system given by the data in Table 2.

Experimental results of friction compensation based on the LuGre model with parameters in Table 2 are shown in Fig. 18. The figure shows the effects of friction which appear as large narrow peaks in the

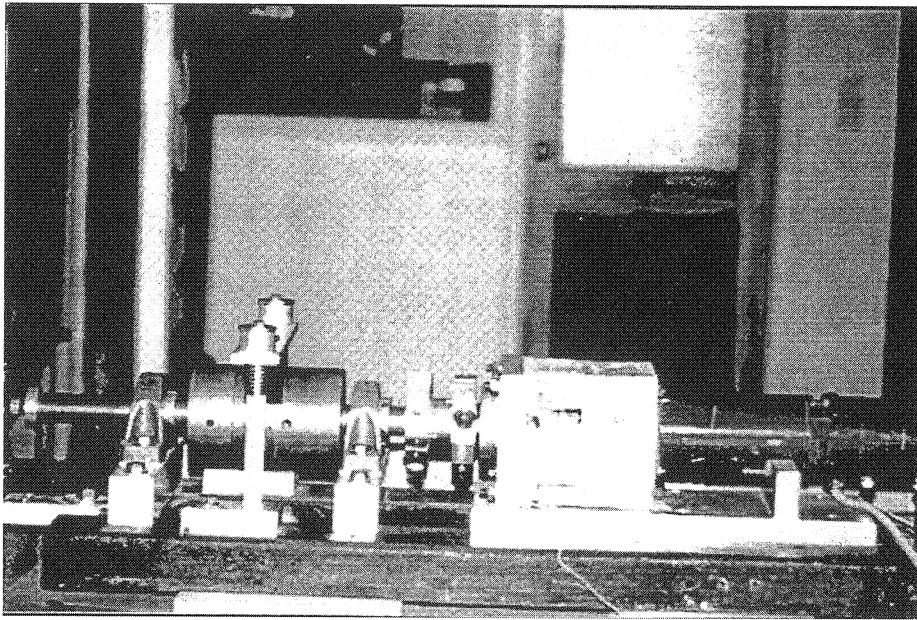


Fig. 17. The test equipment used for experiments with friction compensation.

Table 2. Parameters of the LuGre friction model obtained from system identification. The experiment has been conducted on a DC servo current controller [15]. The Complete model is $J\dot{w} = k_c u = F$, with the total inertia $J = 0.0023$ kg, and voltage/torque constant is $k_c = 1.02$ Nm/V. The friction force is F , described by the LuGre model, with the parameters given below.

Friction parameters	Identified values
α_0	0.28 [N m]
α_1	0.05 [N m]
α_2	0.018 [N m s/rad]
v_0	0.01 [rad/s]
σ_0	260 [N m/rad]
σ_1	0.6 [N m s/rad]

tracking error. The figure also shows the substantial improvements with friction compensation. The peaks in the tracking error practically vanish.

Friction compensation is sensitive to the friction model. This is illustrated in Fig. 19, which shows data from an experiment where a mechanical brake was applied at time $t = 50$ s. The friction compensation, which worked well before, no longer succeeds in reducing the effects of friction. Even if this experiment exaggerates the variations in friction that occur normally it indicates the advantages of adaptive friction compensation.

6.4. Adaptive Friction Compensation

Friction varies with many factors such as normal force, temperature, position, etc. A variation in one of these factors may change the friction characteristics in a complex manner. Since friction depends on so many factors it clearly points to the need for adaptation.

To perform adaptation we must determine which parameters in the model should be adapted. This is a difficult problem because the parameters enter the model in a complicated way. One possibility that has been investigated is based on the model

$$\frac{dz}{dt} = v - \theta \varphi(v, x) \sigma_0 \frac{|v|}{g(v)} z \quad (31)$$

$$F = \sigma_0 z + \sigma_1 \frac{dz}{dt} z + \alpha_2 v \quad (32)$$

where θ describes the unknown parameter vector and $\varphi(v, x)$ the regressor vector. The product $\theta \varphi(v, x)$ captures variations in $1/g(v)$. It is possible to extend the model and make the friction depend on position as well as velocity.

Adaptive observer-based friction compensation mechanisms can be designed for global asymptotic stability (see for example [12]). In the simplest case, only one parameter which represents the magnitude of the friction is adapted. This corresponds to $\varphi(v, x) = 1$ in Eq. (32).

The main difficulty in designing adaptive mech-

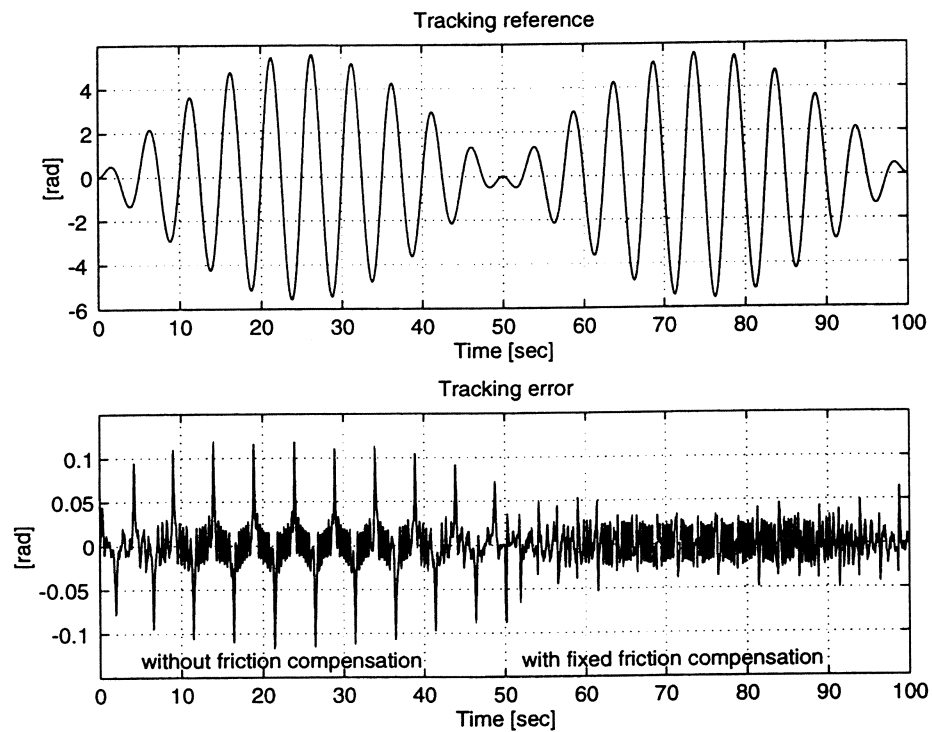


Fig. 18. Tracking experiment with fixed friction compensation based on the LuGre model: reference (upper curve) and position error (lower curve).

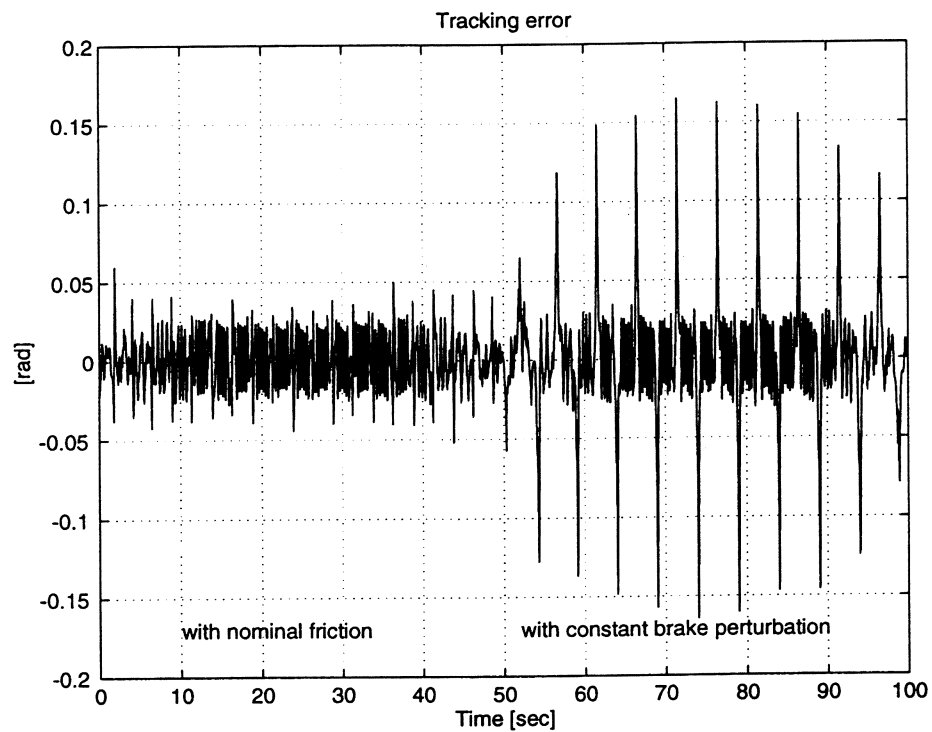


Fig. 19. Tracking experiment with fixed friction compensation: position error under brake perturbation.

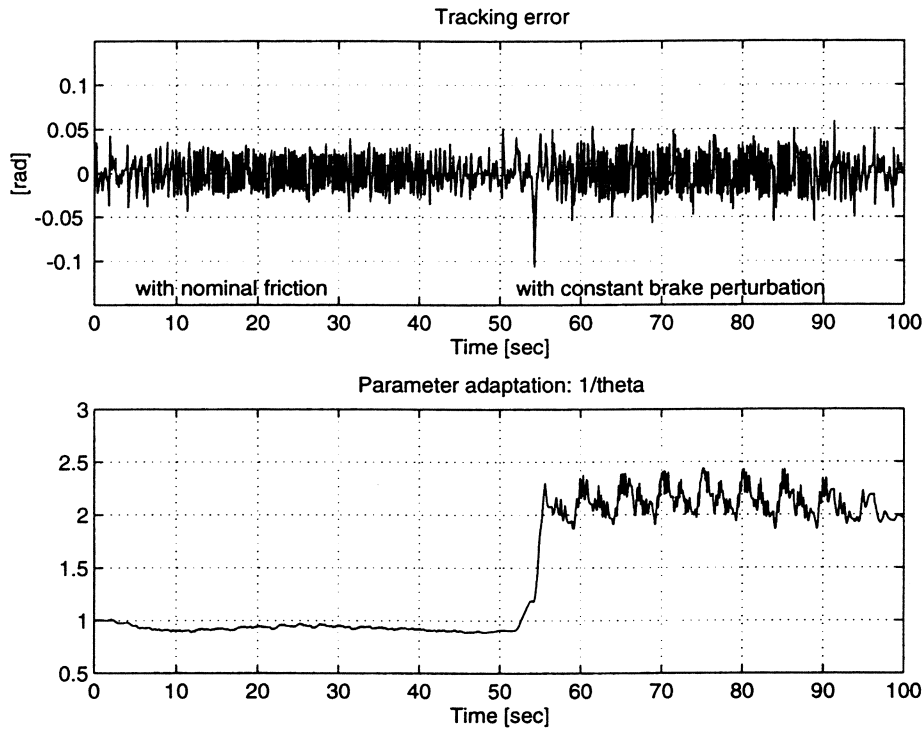


Fig. 20. Tracking experiment with adaptive friction compensation: position error and estimated parameter evolution.

anisms is that the variable z is not measured directly. If the inertia and the nominal friction parameters are assumed to be known, it can, however, be obtained indirectly by filtering the velocity and the control signal. The signal

$$z_m = \frac{1}{\sigma_1 s + \sigma_0} u - \frac{J_s}{\sigma_1 s + \sigma_0} v$$

is related to the true value of z in the following way

$$z = z_m + \mathcal{O}(\exp(-\sigma_0/\sigma_1 t))$$

In the experiments, the signal z_m is used together with the friction observer to drive the adaption by a gradient algorithm.

An advantage of adapting the parameter θ is that it is inversely proportional to the amplitude of normal forces. The adaptation loop automatically gives a way to monitor the variations in the normal force.

An example from Canudas de Wit and Lischinsky [15] is given in Fig. 20. It shows that the adaptive mechanisms manage to give good tracking of the friction parameters after only a few seconds, and that they are able to cope with changes in friction. The adaptive friction compensation has also been applied to a hydraulic industrial robot [36].

7. Conclusions

Friction is present in many control systems for motion control. With the increase in the use of digital control, it is now economically feasible to introduce friction compensation in such systems. This paper has shown the advantage of dynamic friction models over conventional schemes based on static friction models. Present dynamic models such as the LuGre model and the Bliman–Sorine model are reasonably simple and they capture many but not all aspects relevant for friction compensation. Friction compensation based on these models is almost as simple as friction compensation based on static models. Many of the ad hoc fixes traditionally used, e.g. interpolation at low velocities, filtering etc., are also avoided because of the inherent dynamics in the friction model. The results are supported by simulations, experiments in the laboratory and on an industrial hydraulic robot.

Acknowledgements

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